

HDOC - 31

Name	Gurveer Singh Cheema
SAF ID	590032925
Batch	15

Question

Write a C program to perform the following menu-driven operations using separate functions:

1. Get two variable length 2D arrays and prove that matrix addition is commutative.
2. Get two variable length 2D arrays and prove that matrix multiplication is not commutative.
3. Exit

Program

```
#include <stdio.h>

void readMatrix(int r, int c, int a[r][c]) {
    int i, j;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
}

void addMatrix(int r, int c, int a[r][c],
               int b[r][c], int sum[r][c]) {
    int i, j;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            sum[i][j] = a[i][j] + b[i][j];
}

void multiply(int r, int c, int a[r][c],
              int b[r][c], int p[r][c]) {
    int i, j, k;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++) {
            p[i][j] = 0;
            for (k = 0; k < c; k++)
                p[i][j] += a[i][k] * b[k][j];
        }
}

int equal(int r, int c, int a[r][c], int b[r][c]) {
    int i, j;
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            if (a[i][j] != b[i][j]) return 0;
    return 1;
}

int main() {
    int n, choice;
    printf("Enter order of square matrices: ");
    scanf("%d", &n);

    int A[n][n], B[n][n], X[n][n], Y[n][n];

    printf("Enter matrix A:\n");
    readMatrix(n, n, A);
    printf("Enter matrix B:\n");
    readMatrix(n, n, B);

    do {
        printf("\n1. Prove addition is commutative\n");
        printf("2. Prove multiplication is not commutative\n");
        printf("3. Exit\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                addMatrix(n, n, A, B, X);
                addMatrix(n, n, B, A, Y);
                if (equal(n, n, X, Y))
                    printf("A+B = B+A. Addition is commutative.\n");
                break;

            case 2:
                multiply(n, n, A, B, X);
                multiply(n, n, B, A, Y);
                if (!equal(n, n, X, Y))
```

```

        printf("AB != BA. Multiplication is not commutative.\n");
    else
        printf("For these matrices AB = BA.\n");
    break;

    case 3:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice\n");
    }
} while (choice != 3);

return 0;
}

```

Sample Output

Enter order of square matrices: 2

Enter matrix A:

1 2

3 4

Enter matrix B:

5 6

7 8

Enter your choice: 1

A+B = B+A. Addition is commutative.

Enter your choice: 2

AB != BA. Multiplication is not commutative.

Enter your choice: 3

Exiting program...

Result: The program verifies that matrix addition is commutative and demonstrates that matrix multiplication is generally not commutative using separate functions.